Technical Report: Evaluation of Audio Segmentation and Overlap Strategies

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April 21, 2026

1. Introduction

This document evaluates the audio segmentation strategy implemented during the data engineering phase of the underwater vessel noise classification project. By analyzing the `segment_manifest.csv`, we evaluate the mathematical approach used to slice continuous recordings and its subsequent impact on dataset statistics and model generalizability.

2. Segment Length Justification

A fixed **segment length of 3.0 seconds** was utilized. At a standardized sampling rate of 22,050 Hz, the frame count per segment is defined as:

$$N = T \times f_s = 3.0 \times 22,050 = 66,150 \text{ frames} \tag{1}$$

In passive sonar domains, this window length satisfies the wide-sense stationarity assumption. It is sufficiently long to capture full mechanical cycles of low-RPM diesel engines (CargoShip) while remaining short enough to prevent spectral smearing during high-speed maneuvering (SpeedBoat).

3. Overlap and Hop Length Analysis

The hop length utilized during segmentation can be mathematically inferred from the final yield. For the CargoShip class (approx. 6,018 total seconds across 20 files), the contiguous segment count N_C versus the overlapping count N_O is analyzed:

$$N_O = \frac{T_{total} - T_{window}}{T_{hop}} \; 1 \approx 3,980 \text{ segments} \tag{2}$$

Given that N_O is approximately double the yield of a non-overlapping window, this dictates that a **50% overlap (1.5-second hop length)** was employed. This strategy augments training data and captures transient events that might otherwise be split at window boundaries.

4. Statistical Impact: Catastrophic Class Imbalance

While the overlapping strategy maximized data yield, it introduced a severe class imbalance into the pipeline. As illustrated in Table 1, the CargoShip class constitutes **97.1%** of the training set.

Table 1: Segment Distribution Statistics after Standardization

Class	Train Count	Val Count	Test Count	Total Segments
CargoShip	2,786	597	597	**3,980**
KaiYuan	21	4	5	**30**
SpeedBoat	21	5	4	**30**
Uuv	21	4	5	**30**
Noise	21	5	4	**30**
TOTAL	**2,870**	**615**	**615**	**4,100**

5. Comparison with State-of-the-Art Literature

The 3.0-second length aligns with state-of-the-art methodology (Santos-Domínguez et al., 2016) for capturing marine tonal frequencies. A 50% overlap is a widely accepted technique in Environmental Sound Classification (ESC) to augment training data for non-stationary signals.

Final Strategic Recommendations

ACTION REQUIRED: To mitigate the 97% imbalance and prevent model bias, the following interventions are mandated:

1. **Weighted Random Sampler:** Implement undersampling in the dataloader to ensure balanced batch distribution.
2. **Class Penalty:** Apply a weight factor of ≈ 132.6 to minority class loss functions.
3. **Data Augmentation:** Apply time-masking and pitch-shifting to minority classes to artificially inflate representation.